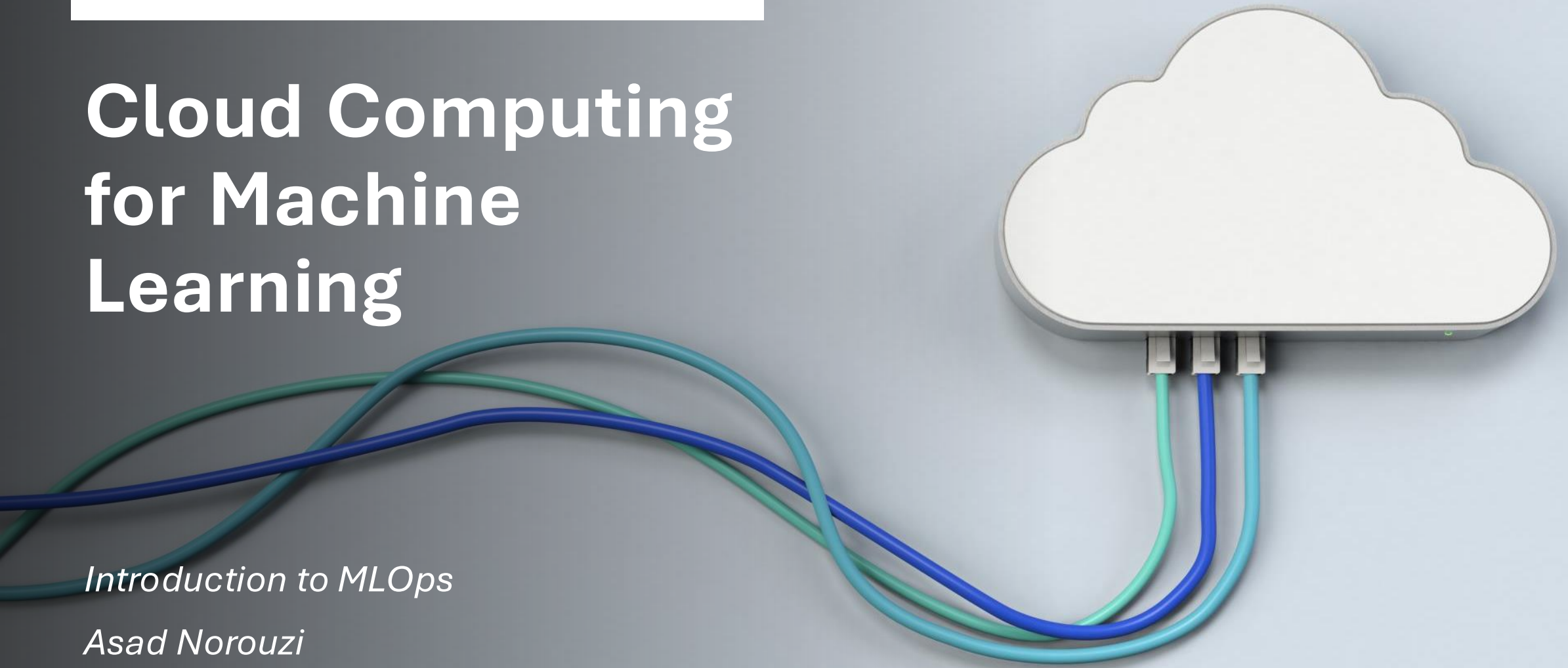

Cloud Computing for Machine Learning

Introduction to MLOps

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Agenda

What is MLOps?

The Need for MLOps

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What is MLOps?

MLOps, or Machine Learning Operations, is a set of practices that aims to deploy and maintain machine learning models in production reliably and efficiently.

The goal of MLOps is to streamline the end-to-end machine learning development process, including integration, testing, releasing, deployment, and infrastructure management.

MLOps facilitates collaboration between ML scientists and operations professionals to ensure that machine learning models are scalable, reproducible, and sustainable.

The Need for MLOps

Machine learning models face unique challenges in production, such as model drift, data quality issues, and scalability problems.

MLOps addresses these challenges by applying DevOps principles to machine learning, ensuring models are updated, reliable, and delivering value.

MLOps reduces time to market, improves model performance, and enhances team collaboration.

Key Components of MLOps

Version Control: For models, data, and code to track changes and collaborate.

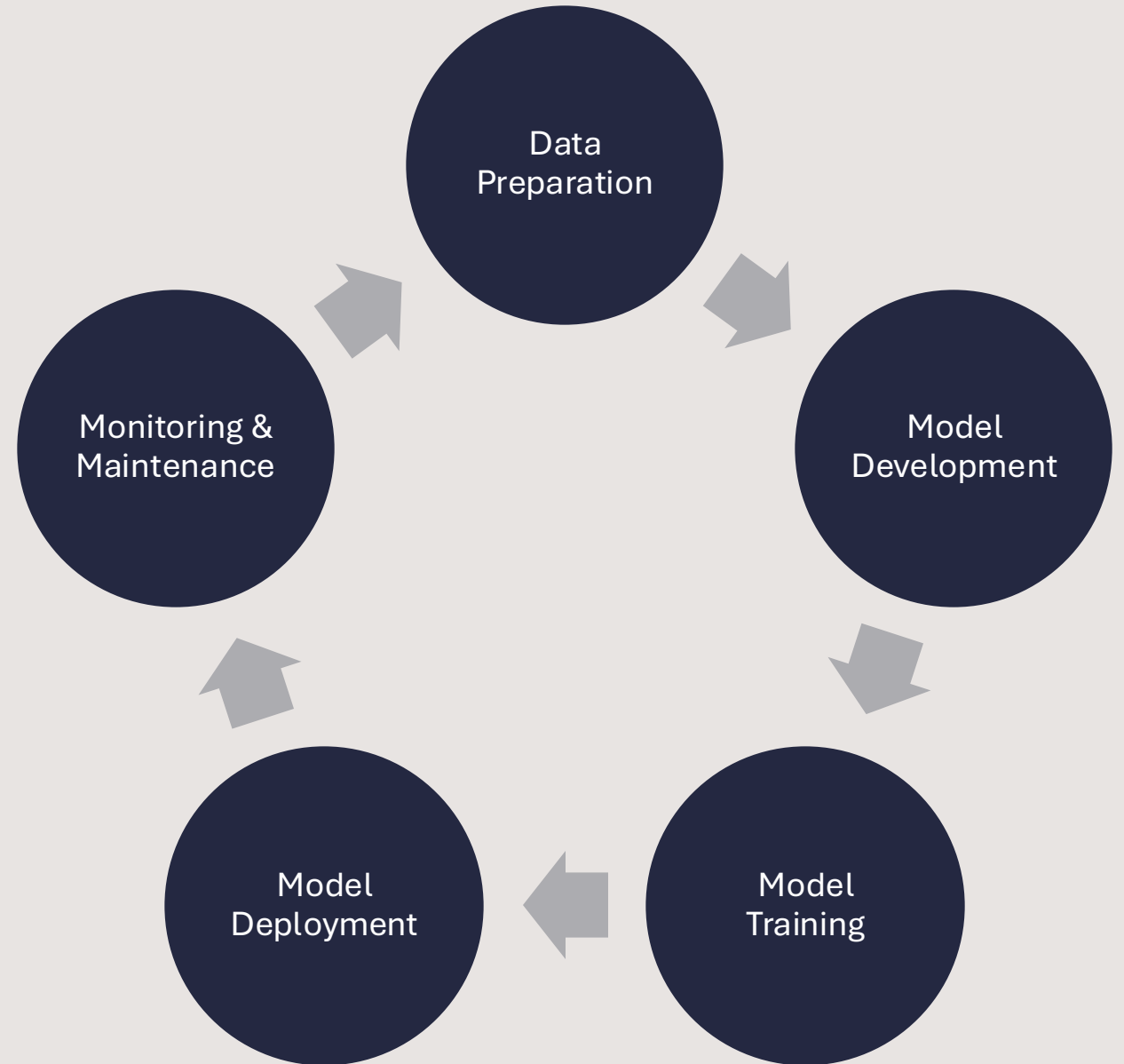
Continuous Integration and Continuous Delivery (CI/CD): For automating the testing and deployment of models.

Model Monitoring: To track performance and identify when retraining is necessary.

Data Management: Ensuring data quality and accessibility for training and inference.

Model Governance: Establishing standards and ethics guidelines for model development and use.

MLOps Lifecycle



Tools & Technologies in MLOps

Machine Learning Frameworks:
TensorFlow, PyTorch, Scikit-learn.

MLOps Platforms: MLflow, Kubeflow,
TFX.

Containerization: Docker, Kubernetes
for scalable deployment.

Monitoring Tools: Prometheus,
Grafana.

Workflow Automation: Jenkins,
GitHub, GitLab CI/CD.

MLOps Best Practices

Automation: Automate as much of the ML lifecycle as possible to reduce errors and speed up delivery.

Monitoring: Continuously monitor model performance to detect and correct drift.

Collaboration: Foster a culture of collaboration between data scientists, ML engineers, and DevOps teams.

Scalability: Design systems with scalability in mind to handle increased loads and data volumes.

Ethics and Governance: Incorporate ethical considerations and governance from the outset.

Summary

MLOps is critical for deploying, maintaining, and scaling machine learning models in production.

Continuous evolution with advances in AI and cloud technologies.



Q&A
